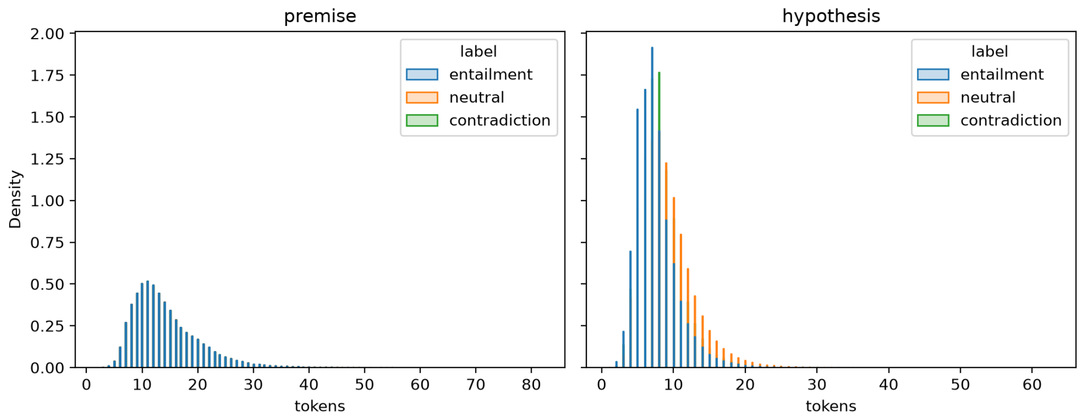
NLP1 Project 1 — NLI on SNLI

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github.com/neelshahx/nlp1-2026

Q1 · Task & data

- (premise, hypothesis) → entailment / neutral / contradiction
- SNLI: 549k train / 9.8k dev / 9.8k test, balanced (~33% each)
- Artifacts (Gururangan et al. 2018): negation → contradiction, longer neutral hypotheses
- Token overlap (Jaccard): ent 0.31 · neu 0.26 · con 0.22
- PMI(token, label) = log p(token | label) / p(token)
- Setup: stratified 5-fold CV on train (seed 0); dev for selection; test once



Q2 · Tokenizer

- BPE (HF tokenizers), min_freq 2, trained on SNLI train (premise + hypothesis) only
- 549k pairs is enough data: no external corpus needed
- Vocab 4k / 8k / 16k → CV acc 64.0 / 64.2 / 63.9: flat
- Word-level baseline 63.9 vs BPE 64.3 → inside ROPE; BPE wins on coverage, not accuracy
- Subwords → 0% out-of-vocabulary on dev (word-level 0.2–0.6%)
- Caveat: fit on all of train (incl. held-out fold), but no labels used

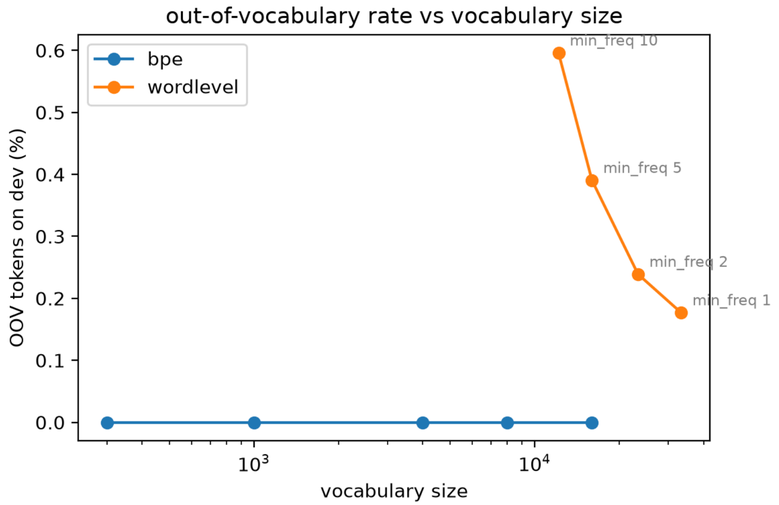
first merges learned (SNLI train)

```
 1.  i + n  →  in
 2.  in + g_  →  ing_
 3.  t + h  →  th
 4.  a + n_  →  an_
 5.  a + r  →  ar
 6.  th + e_  →  the_
 7.  a + n  →  an
 8.  i + s_  →  is_
 9.  i + n_  →  in_
10.  m + an_  →  man_
11.  i + t  →  it
12.  r + e  →  re
```

"Two kids at a ballgame wash their hands."

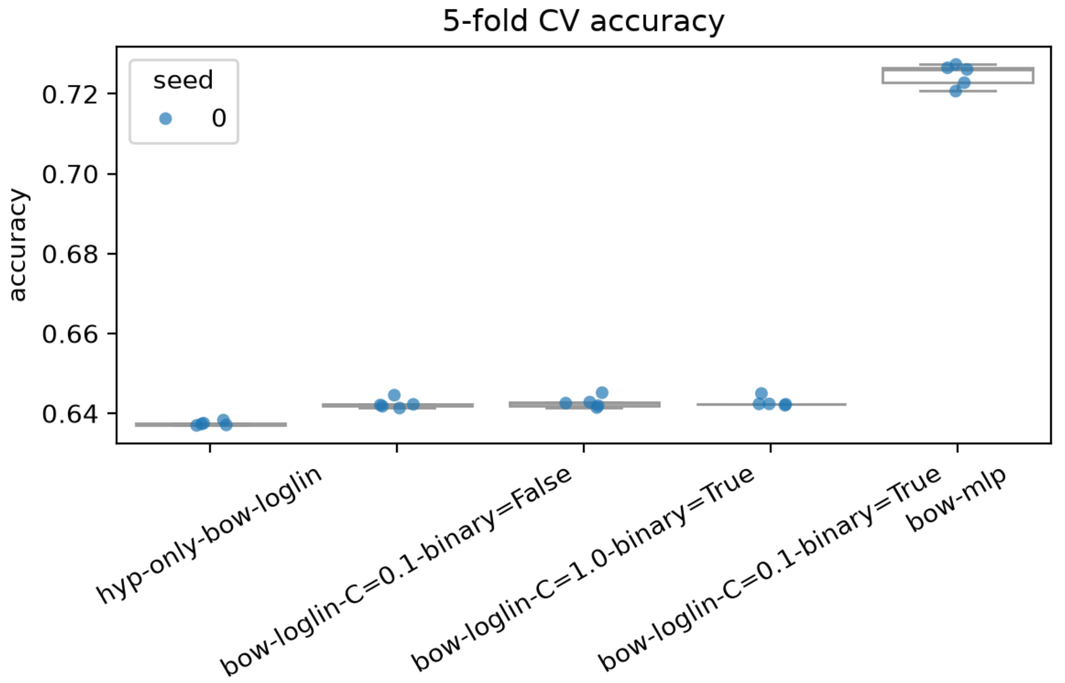
vocab 300 (21 tokens)	two · k · id · s · at · a · b · al · l · g · am · e · w · as · h · the · ir · h · an · ds · .
vocab 1000 (13 tokens)	two · kids · at · a · bal · l · game · w · as · h · their · hands · .
vocab 4000 (11 tokens)	two · kids · at · a · ball · game · w · ash · their · hands · .
vocab 8000 (10 tokens)	two · kids · at · a · ball · game · wash · their · hands · .
vocab 16000 (9 tokens)	two · kids · at · a · ballgame · wash · their · hands · .

blue = ends a word, orange = word continues



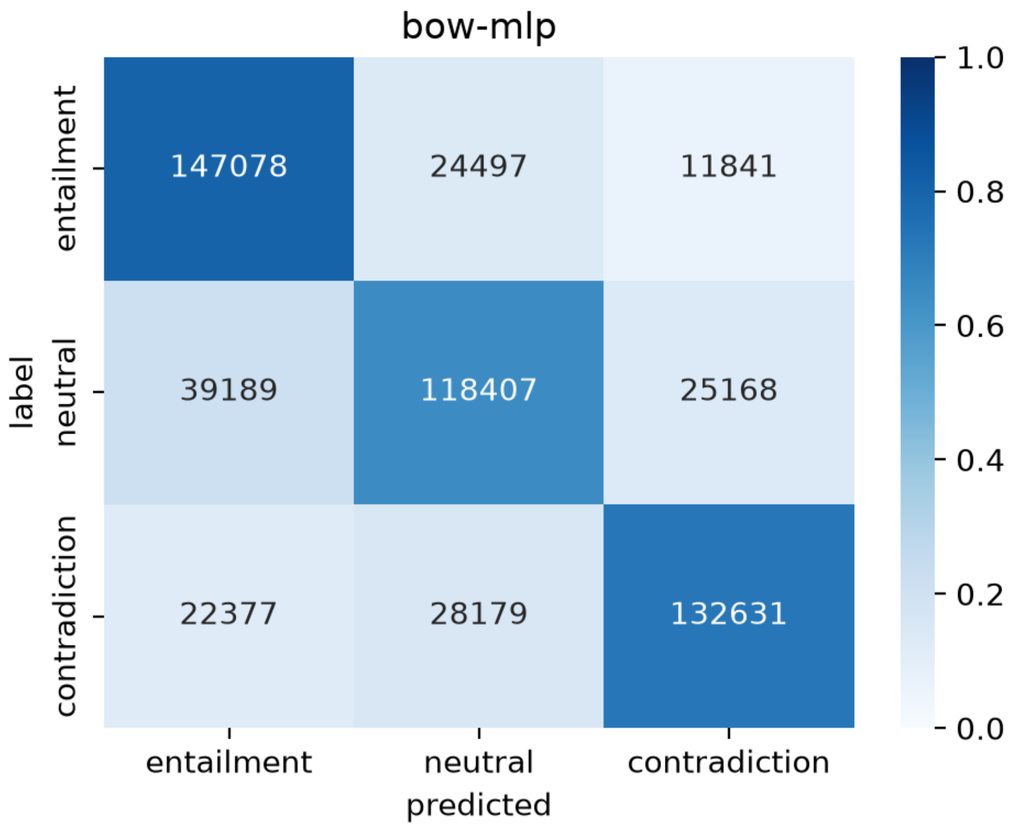
Q3 · Representation

- Both sides at once: concat [BoW(P) ; BoW(H)], so the model knows which side a token is on
- Joint (one bag over "P [SEP] H" = BoW(P) + BoW(H)) 58.8 vs concat 64.3: –5.5 pt, side information lost
- Hypothesis-only 63.7 vs concat 64.2, P(rope) = 0.997
- Indicators vs counts: 64.28 vs 64.24 → inside ROPE (short sentences)
- Linear on concat = f(P) + g(H): no P × H interaction
- n-grams 1–2 (local word order): unigrams 64.2, bigrams TBD → next: TF-IDF, cross features |u – v|, u ⊙ v



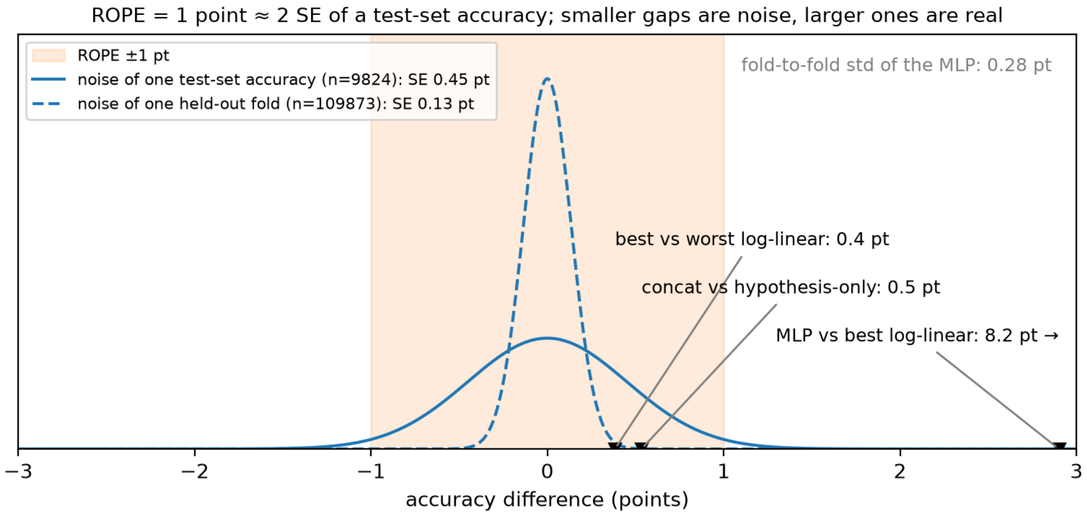
Q4 · Architectures

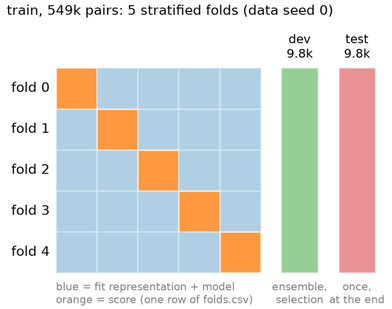
- Log-linear (lbfgs, L2), C ∈ {0.1, 1, 10}: flat ≈ 64%
- MLP: 16k → 128 → ReLU → dropout 0.1 → 3
- Adam 1e-3, weight decay 1e-4, early stopping (10% val, patience 3)
- MLP 72.5% (+8 pts): hidden layer mixes P and H
- Third: residual MLP (LayerNorm + skip), blocks {0, 2}, hidden {128, 512} — TBD

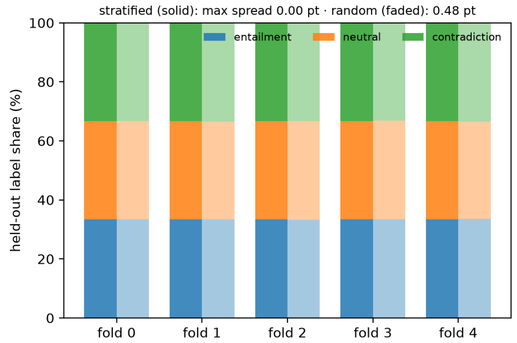


Q5 · Evaluation

- Stratified 5-fold CV, same folds for all → paired; equal label share per fold (random split: ±0.5 pt)
- Accuracy (+ macro-F1; balanced, so ≈ equal)
- Floors: majority 33%, hypothesis-only 63.7%
- Correlated Bayesian t-test (baycomp, ρ = 1/k)
- ROPE = 1 accuracy point ≈ 2 SE of a test-set accuracy (SE ≈ 0.45 pt)
- Neutral is the hardest class







Q6 · Best model

- BoW + MLP: CV 72.5 ± 0.3
- P(MLP > log-linear) = 1.00, P(rope) = 0
- Log-linear variants all inside ROPE → pick simplest
- Dev, 5-fold ensemble: MLP 77.6 vs log-linear 66.5
- Test: once, final model only — TBD

